SDG-DSB: Spectral Degradation Guided Diffusion Schrödinger Bridge for Hyperspectral Images Super-Resolution

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***Abstract*—Hyperspectral image (HSI) super-resolution plays a crucial role in enhancing the spatial resolution of HSIs while preserving spectral integrity. A common approach for HSI super-resolution involves fusing low-resolution HSIs (LR-HSIs) with high-resolution multispectral images (HR-MSIs). In this article, we propose a novel spectral degradation guided Diffusion Schrödinger Bridge (SDG-DSB) model specifically designed for HSI super-resolution. Recognizing that LR-HSIs contain complete spectral information, we leverage a subspace model to decompose LR-HSIs into endmembers and LR abundance maps. To reconstruct the HR-HSI, we propose a DSB, transforming LR abundance maps into HR counterparts. This process is supported by multiple pretrained DSB generators trained on natural images, integrating spatial priors for improved accuracy. Additionally, the HR-MSI guides the diffusion process, maintaining consistency with the spectral degradation path from HR-HSI to HR-MSI. This spectral degradation captures the inherent correlations between spectral bands, allowing for a more accurate restoration of high-frequency spatial details while maintaining the integrity of spectral information. Finally, the HR abundance maps are combined with endmembers extracted from LR-HSI to generate the HR-HSI. Extensive experiments on benchmark HSI datasets show that the proposed SDG-DSB outperforms state-of-the-art methods, demonstrating its effectiveness in both spatial and spectral reconstruction.**

***Index Terms*—Diffusion Schrödinger bridge (DSB), hyperspectral images (HSIs), multispectral images (MSIs), super-resolution.**

I. Introduction

HYPERSPECTRAL imaging captures images across hundreds or even thousands of narrow, contiguous spectral bands, providing detailed spectral information for each pixel. The high spectral resolution in hyperspectral images (HSIs) enables precise material identification and differentiation [1], [2]. However, HSIs often suffer from low spatial resolution due to sensor limitations and the inherent tradeoff between spatial and spectral resolution. Given the high dimensionality of spectral data in HSIs, enhancing spatial resolution in a manner similar to single-image super-resolution for natural images presents significant challenges.

Conversely, multispectral imaging captures a smaller number of broader spectral bands—typically three to ten—and can produce high spatial resolution multispectral images (HR-MSIs). A common and increasingly popular approach for HSI super-resolution involves fusing low-resolution HSI (LR-HSI) with high-resolution multispectral image (HR-MSI). This fusion leverages the rich spectral detail of HSIs and the high spatial resolution of MSIs, producing an enhanced image that maximizes the strengths of both [3], [4], [5], [6]. This fused imagery enables more accurate and detailed information for various remote sensing applications, enhancing the performance of tasks such as land cover classification, target detection, and environmental monitoring. For brevity, we call the HSI and MSI fusion-based super-resolution as HSI super-resolution.

Existing methods for HSI super-resolution can be broadly categorized into three approaches: detail-injection-based methods, variational optimization-based methods, and deep learning (DL)-based methods. Detail-injection-based methods involve extracting spatial details from HR-MSIs and injecting them into the fused images [7], [8]. Variational optimization-based methods first represent the HR-HSI using matrix or tensor factorization models [9], [10]. DL-based methods have gained significant attention due to the rapid advancements in DL technology. In these methods, a mapping from LR-HSI and HR-MSI to HR-HSI is learned using deep convolutional neural networks (CNNs) [11], [12], [13], [14], [15] or transformer networks [16], [17]. However, DL-based methods are fully data-driven and typically do not incorporate prior knowledge, which can limit their robustness and generalization.

Generally speaking, the HSI super-resolution is a typical inverse problem, where the goal is to recover the underlying signals or images from incomplete, noisy, or indirect measurements. Inverse problems are often ill-posed, meaning that they lack a unique solution or are highly sensitive to

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measurement errors. Recently, generative models have achieved great success in a wide range of applications, from natural language processing to image synthesis and beyond. In particular, generative models have proven to be highly effective in inverse problems. By leveraging large datasets, these models learn complex priors about the data, which can then be used to guide the reconstruction process in inverse problems. This data-driven approach contrasts with traditional methods that rely on handcrafted priors and optimization techniques.

Moreover, generative models can incorporate physical models and constraints directly into the learning process, further enhancing their performance in inverse problems. Techniques such as variational autoencoders (VAEs) [18], generative adversarial networks (GANs) [19], and diffusion models [20], [21], [22] have been particularly successful, as they can model complex data distributions and produce outputs that adhere to the underlying physical principles of the problem at hand.

Although the diffusion-based methods have achieved promising results in HSI super-resolution, these methods can only generate the HR-HSI from Gaussian noise, which does not reflect the real super-resolution process. In this article, we intend to introduce a novel framework by generating the HR image from certain observations. The proposed method involves finding the most likely stochastic process that bridges two given probability distributions over time called Schrödinger Bridge [23], [24]. Specifically, given an initial and a final distribution, the Schrödinger Bridge seeks to identify the most probable evolution of the system's state, described by a diffusion process, that transforms the initial distribution into the final one. In this article, we propose the diffusion Schrödinger bridge (DSB) to describe the evolution of HR-HSI, ensuring that the transformation from LR-HSI to HR-HSI is both optimal and consistent with physical constraints. However, the generated image only satisfies the probability distribution of the desired dataset. The data consistency with respect to the given observation is not ensured. To solve this problem, the diffusion posterior sampling (DPS) [25] method has been proposed by modifying the sampling algorithm under the Bayesian rule. The main idea is to constantly guide the diffusion trajectory [26] satisfying the data degradation. Different from other inverse problems, in the HSI super-resolution, we have two observations, which result in two data consistency terms. Some methods roughly combine the two data consistency terms with a weight. However, the weight is a free parameter that is hard to determine. In this article, we generate the HR-HSI from the LR-HSI and introduce the spectral degradation guidance in the diffusion process. By this approach, we can avoid the weight which has great impact on the HR results.

Meanwhile, the training of the DSB is challenging. First, the optimization process must balance multiple objectives, including minimizing the difference between the generated output and the target, while also satisfying an entropy regularization term that ensures smooth and physically consistent transitions. Moreover, the DSB involves a bidirectional process, where both forward (generation) and backward (degradation) paths need to be trained to ensure consistency. This bidirectionality adds to the computational complexity, as it requires careful tuning to maintain coherence between the two processes. Additionally, in real applications, we do not have enough training data (i.e., paired LR-HSI, HR-MSI, and HR-HSI). Luckily, we have the pretrained models trained on a large amount of natural images. However, these pretrained models only generate three-band images, which is not the case of HSI. To solve this problem, we propose a multiple generation strategy to generate multiple three-band images and stack these three-band images along the spectral dimension as the abundance image. Then, by multiplying with the spectral endmembers estimated from the LR-HSI, we can obtain the desired HR-HSI.

Moreover, diffusion models are employed for HSI super-resolution due to a compelling alignment between their inherent strengths and the core challenges of the task. First, diffusion models learn powerful generative priors from data, enabling them to synthesize highly realistic spatial textures—a significant advantage over hand-crafted priors for this severely ill-posed problem. Second, their iterative refinement process provides a natural framework for solving inverse problems. The step-by-step denoising can be systematically guided by physics-based degradation models, ensuring the reconstruction remains consistent with all observed inputs. Finally, this guided, progressive generation is instrumental in preserving spectral fidelity. By gradually enforcing consistency with the spectral data, the model avoids the spectral distortions common in other generative approaches, ensuring the integrity of the reconstructed hyperspectral data. This combination of a strong prior, flexible iterative refinement, and inherent suitability for spectral preservation makes diffusion models particularly well-suited for HSI super-resolution.

In summary, the highlights of the proposed method are threefold.

1) The diffusion Schrödinger bridge is proposed for HSI super-resolution. The LR-HSI is used to generate the HR-HSI and the HR-MSI is leveraged as the spectral degradation guidance in the diffusion process. It is the first time that spectral degradation guidance is introduced in the DSB.
2) To make full use of the spatial prior of the spatial information, a multiple generation strategy, which employs multiple pretrained Schrödinger bridge generators trained with natural images to generate a high spatial resolution abundance is proposed. In this way, we can avoid the difficult training issue of the DSB and the shortage of training images.
3) We compare our method with state-of-the-art methods including model-based and DL-based methods. Experimental results on four datasets demonstrate the advantages of the proposed method.

The remainder of the article is organized as follows. Section II describes the related work. In Section III, the proposed method for HSI super-resolution is introduced in detail. Section IV conducts extensive experiments on HSI datasets to demonstrate the advantages of the proposed method. Finally, conclusion is drawn in Section V.

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II. Related Work

In this section, we will present the related HSI super-resolution works and then give a brief introduction to the inverse problems with diffusion models.

A. HSI Super-Resolution

HSI offers rich spectral information but often suffers from low spatial resolution. To improve the spatial resolution of HSI, HR remote sensing images, such as panchromatic (PAN), RGB, or other MSIs, are used to provide spatial information, a process referred to as HSI-MSI fusion for super-resolution. In this article, we focus on combining LR-HSI with HR-MSI or similar images to achieve fusion-based super-resolution.

Detail-injection methods replace one component of HSI with HR images. For example, intensity–hue–saturation (IHS) [27], [28] substitutes the intensity channel of the LR image with a PAN image. Similarly, principal component analysis (PCA) [29] replaces the first principal component of LR-HSI with a HR image. Although effective at enhancing spatial resolution, these approaches often introduce spectral distortions in the reconstructed HR-HSI.

Variational optimization-based methods build models to characterize HR-HSI and estimate unknown variables using the observed LR-HSI and HR-MSI. Yokoya et al. [30] introduced the coupled nonnegative matrix factorization (CNMF) model, which decomposes HR-HSI into endmembers and abundance maps. Dong et al. [31] introduced a nonnegative structured sparse representation model that estimates spectral dictionaries and sparse coding coefficients simultaneously. Other methods, such as [32], utilize prior information, while tensor-based models [9], [10], [33], [34] focus on preserving the multidimensional structure of HR-HSI.

DL-based methods learn mappings from LR-HSI and HR-MSI to HR-HSI using the observed data. For instance, Yang et al. [35] proposed a dual-branch CNN to extract information separately, while Zhang et al. [36] used a physical model to maintain spectral structures. Other notable works include [37], [38], [39], and [40], which employed CNN denoisers, deep image prior, and deep blind super-resolution models. Cao et al. [41] proposed an unsupervised multilevel spatio-spectral fusion transformer (UMSFT) for HSI super-resolution. This method was designed with a multilevel transformer architecture for explicit spatio-spectral feature fusion, which achieves promising results with unsupervised training. Further, cross-domain-aware transformer (CaFormer) [42], a deep unrolling network that represents a significant evolution within the optimization-inspired paradigm for HSI super-resolution. However, these methods are often limited by the need to train specific models for each observation, which restricts their generalization ability.

Recently, diffusion models have also been applied to HSI-MSI fusion. Li et al. [43] proposed a dual conditional diffusion model-based fusion network (DCDMF) that uses two noise prediction networks with different conditional inputs. In [44], a low-rank diffusion model was employed for HSI pansharpening, utilizing a pretrained diffusion model as a prior under the Bayesian framework. Additionally, Qu et al. [45] introduced a spatial–spectral-bilateral cycle-diffusion framework (S^2CycleDiff) to generate HR-HSI step-by-step through a customized conditional cycle-diffusion network.

B. Inverse Problems With Diffusion Model

Generative models have gained significant attention in computer vision, especially for multimodal generation. Among these, diffusion models have shown outstanding performance in image generation by learning the underlying data distribution. These models are often pretrained on large-scale image datasets, allowing images to be recovered in a plug-and-play fashion that incorporates the measurement process. The denoising diffusion restoration model (DDRM) [46] was the first to use a pretrained diffusion model as a prior, motivated by variational inference. Later, Song et al. [22] enhanced this by projecting the image onto the measurement subspace to improve the likelihood score, although this approach struggles with noisy measurements.

The denoising diffusion null-space model (DDNM) [47] addresses arbitrary linear inverse problems, while DPS [25] uses the Laplace approximation to estimate the distribution of the original data, improving on diffusion-based inverse methods. A parallel diffusion model was later proposed by Chung et al. [25], enabling the processing of multiple variables simultaneously via parallel reverse diffusion. To solve HSI restoration, a self-supervised diffusion model called DDS2M [48] was introduced, generating spectral and spatial subspaces through diffusion sampling. Although it delivered promising results, its networks were randomly initialized, limiting its ability to leverage prior HSI distributions.

In an attempt to address the lack of HSI-specific priors, an unsupervised HSI pansharpening method based on a low-rank diffusion model [44] was proposed. While this method used a pretrained model to generate low-rank representations, the model was trained on natural images and used fixed parameters, which may not be optimal for HSI applications. Moreover, the lack of explainability further restricts its use.

The strength of diffusion models lies in their ability to learn data distributions through well-trained networks. However, training these networks requires massive datasets due to the high-dimensionality of signals, which is impractical in many real-world applications. Therefore, more efficient and easily trainable networks need to be explored for broader applicability.

In addition, these diffusion-based super-resolution methods often start from Gaussian noise to generate the clean image. This makes it difficult to accurately model the true generation process of the HR-HSI. To overcome these limitations, we propose the spectral degradation guided DSB (SDG-DSB) method, which starts from the LR-HSI and introduces HR-MSI as the spectral degradation guide. Meanwhile, it minimizes the need for extensive model training and additional data, making it more practical for real-world applications.

III. Proposed Method

A. HSI Super-Resolution

The problem of HSI super-resolution can be formulated as follows.

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Let:

1) $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$ be the HR-HSI, where H and W are the spatial dimensions, and L is the number of spectral bands.
2) $\mathbf{Y} \in \mathbb{R}^{h\times w\times L}$ be the LR-HSI, where $h < H$ and $w < W$.
3) $\mathbf{Z} \in \mathbb{R}^{H\times W\times l}$ be the HR-MSI, where $l < L$.

The goal of HSI super-resolution is to estimate the HR-HSI $\hat{\mathbf{X}}$ from $\mathbf{Y}$ and $\mathbf{Z}$.

The degradation process can be described by the following formulas:

$$\mathbf{Y} = \mathbf{X} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2 + \mathbf{N}_{\text{HSI}} \tag{1}$$

$$\mathbf{Z} = \mathbf{X} \times_3 \mathbf{R} + \mathbf{N}_{\text{MSI}} \tag{2}$$

where $\mathbf{P}_1 \in \mathbb{R}^{H\times h}$ and $\mathbf{P}_2 \in \mathbb{R}^{W\times w}$ are the spatial degradation matrices, modeling the spatial downsampling from HR to LR. $\mathbf{R} \in \mathbb{R}^{L\times l}$ is the spectral response function (SRF), modeling the spectral degradation from HR-HSI to HR-MSI. $\mathbf{N}_{\text{HSI}}$ and $\mathbf{N}_{\text{MSI}}$ are noise terms associated with the HSI and MSI, respectively.

Although we have two observations, the number of parameters is much less than the parameters in the HR-HSI. To obtain our desired result, additional regularization is introduced in the super-resolution method. Then, the objective is to recover $\mathbf{X}$ by solving the inverse problem

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \left\|\mathbf{Y} - \mathbf{X} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\right\|_F^2 + \lambda\left\|\mathbf{Z} - \mathbf{X} \times_3 \mathbf{R}\right\|_F^2 + \phi(\mathbf{X}) \tag{3}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. λ is a regularization parameter that balances the contributions of the two data fidelity terms. $\phi(\mathbf{X})$ is a regularization term that imposes prior knowledge or constraints on $\mathbf{X}$.

B. DSB HSI Super-Resolution

Diffusion models represent a powerful category of generative models that learn to reverse a diffusion process, mapping data to noise [21], [22]. Despite their impressive performance, standard diffusion models are not always well-suited for tasks where the prior distribution is not simply random noise. For instance, in image translation tasks, the goal is to establish a mapping between pairs of images, which is challenging within the conventional diffusion framework.

The DSB provides a robust mathematical framework designed to interpolate between two probability distributions over a finite time horizon. Grounded in stochastic processes and optimal transport principles, the DSB seeks to identify a stochastic process governed by a diffusion equation that ensures its marginal distributions match the specified initial and final distributions. Conceptually, the DSB can be framed as a regularized optimal transport problem, where the objective is to minimize the Kullback–Leibler divergence between the probability flow and a reference process.

We illustrate the DSB and its data-consistent variant using paired LR-HSI $\mathbf{Y}$ and HR-HSI $\mathbf{X}$ pairs. Following the formulation presented in I^{2}SB [23], we assume that $\mathbf{X}_0 := \mathbf{X} \sim p(\mathbf{X})$ and $\mathbf{X}_1 := \mathbf{Y} \sim p(\mathbf{Y} \mid \mathbf{X})$. The posterior distribution of $\mathbf{X}_t$ is defined as the product of two Gaussian distributions

$$p(\mathbf{X}_t \mid \mathbf{X}_0, \mathbf{X}_1) = \mathcal{N}\left(\mathbf{X}_t; \frac{\bar{\gamma}_t^2}{\gamma_t^2+\bar{\gamma}_t^2}\mathbf{X}_0 + \frac{\gamma_t^2}{\gamma_t^2+\bar{\gamma}_t^2}\mathbf{X}_1, \frac{\gamma_t^2\bar{\gamma}_t^2}{\gamma_t^2+\bar{\gamma}_t^2}\boldsymbol{I}\right). \tag{4}$$

We can sample $\mathbf{X}_t$ using the reparameterization trick

$$\mathbf{X}_t = (1-\alpha_t)\mathbf{X}_0 + \alpha_t\mathbf{X}_1 + \sigma_t\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{I}) \tag{5}$$

where $\alpha_t := (\gamma_t^2)/(\gamma_t^2+\bar{\gamma}_t^2)$ and $\sigma_t^2 := (\gamma_t^2\bar{\gamma}_t^2)/(\gamma_t^2+\bar{\gamma}_t^2)$. The DSB can be interpreted as a denoising process that transitions from a clean image at $t = 0$ to a state of maximum degradation at $t = 1$ through a convex combination of $(\mathbf{X}_0, \mathbf{X}_1)$.

This diffusion bridge introduces a continuous degradation process by blending $(\mathbf{X}_0, \mathbf{X}_1)$ while incorporating stochastic elements via the noise term σ_t. Our objective is to train a time-conditioned neural network to map $\mathbf{X}_t$ back to $\mathbf{X}_0$ for any time step t, thereby recovering the original HR image. The training loss of I^{2}SB [23], akin to denoising score matching (DSM) [22], is given by

$$\min_\theta \mathbb{E}_{\mathbf{Y}\sim p(\mathbf{Y}|\mathbf{X}),\mathbf{X}\sim p(\mathbf{X}),t\sim U(0,1)}\left[\left\|s_\theta(\mathbf{X}_t) - \frac{\mathbf{X}_t - \mathbf{X}_0}{\gamma_t}\right\|_2^2\right]. \tag{6}$$

This can also be formulated as training a residual network H_θ designed for HSI

$$\min_\theta \mathbb{E}\left[\|H_\theta(\mathbf{X}_t) - \mathbf{X}_0\|_2^2\right]. \tag{7}$$

For simplicity, we denote the trained network as H_{θ^*}. Once trained, we can reconstruct $\mathbf{X}_0$ starting from $\mathbf{X}_1$ using methods such as DDPM ancestral sampling [21]. For $s < t$, the posterior distribution is expressed as

$$p(\mathbf{X}_s \mid \mathbf{X}_0, \mathbf{X}_t) = \mathcal{N}\left(\mathbf{X}_s; \left(1-\alpha_{s|t}^2\right)\mathbf{X}_0 + \alpha_{s|t}^2\mathbf{X}_t, \sigma_{s|t}^2\boldsymbol{I}\right) \tag{8}$$

where

$$\alpha_{s|t}^2 := \frac{\gamma_s^2}{\gamma_t^2}, \quad \sigma_{s|t}^2 := \frac{\left(\gamma_t^2-\gamma_s^2\right)\gamma_s^2}{\gamma_t^2}. \tag{9}$$

At inference, $\mathbf{X}_0$ is estimated with $\hat{\mathbf{X}}_{0|t}$ through the trained network to obtain $\mathbf{X}_s \sim p(\mathbf{X}_s \mid \hat{\mathbf{X}}_{0|t}, \mathbf{X}_t)$.

While the reverse diffusion process initiates directly from the observation (or its equivalent), the estimates $\hat{\mathbf{X}}_{0|t} = H_\theta(\mathbf{X}_t)$ are often imperfect, which does not always guarantee the preservation of the observation condition $\mathbf{Y} \approx \mathbf{X} \times {}_1\mathbf{P}_1 \times {}_2\mathbf{P}_2$. Consequently, the sampling trajectory can easily deviate from the path, resulting in an increase in the residual error and leading to diminished sample quality, particularly in terms of distortion.

To solve this issue, one effective approach is to maintain the direct diffusion bridges [23] sampling strategy while augmenting it with additional steps that consistently guide the trajectory to satisfy data consistency. This method is analogous to the gradient guidance used in inverse problems. Since clean image estimates $\hat{\mathbf{X}}_{0|t}$ are generated at each iteration, the residual with respect to the observation $\mathbf{Y}$ can be utilized. The data consistent DDB (CDDB) [49] method incorporates a gradient adjustment that minimizes this residual after each sampling step.

Rewriting (8) with the reparameterization trick, we have

$$\mathbf{X}_s = \hat{\mathbf{X}}_{0|t} + \alpha_{s|t}^2\left(\mathbf{X}_t - \hat{\mathbf{X}}_{0|t}\right) + \sigma_{s|t}\boldsymbol{\epsilon}, \quad \hat{\mathbf{X}}_{0|t} := H_{\theta^*}(\mathbf{X}_t). \tag{10}$$

In DDIM [50], it is claimed that if the noise estimation network is powerful and well expressive, the noise term

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Algorithm 1 DSB HSI Super-Resolution

Require: H_{θ^*}, LR-HSI $\mathbf{Y}$, α_t, σ_t, $\alpha^2_{t-1|t}$, $\sigma^2_{t-1|t}$, ρ

1: Initialize $\mathbf{X}_N$ = upsample($\mathbf{Y}$) using bicubic upsampling method
2: **for** $t = N$ **to** 1 **do**
3: $\quad\hat{\mathbf{X}}_{0|t} \leftarrow H_{\theta^*}(\mathbf{X}_t)$
4: $\quad\mathbf{X}'_{t-1} \leftarrow (1-\alpha^2_{t-1|t})\hat{\mathbf{X}}_{0|t} + \alpha^2_{t-1|t}\mathbf{X}_t + \sigma_{t-1|t}\boldsymbol{\epsilon}$
5: $\quad\mathbf{g}_t \leftarrow \nabla_{\mathbf{X}_t}\|\mathbf{Y} - \hat{\mathbf{X}}_{0|t} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\|_2^2$
6: $\quad\mathbf{X}_{t-1} \leftarrow \mathbf{X}'_{t-1} + \rho\mathbf{g}_t$
7: **end for**
8: **return** $\mathbf{X}_0$

$\mathbf{X}_t - \hat{\mathbf{X}}_{0|t}$ becomes Gaussian, satisfying the total variance condition

$$\left(\alpha_{s|t}\sigma_t\right)^2 + \sigma^2_{s|t} = \sigma^2_s \tag{11}$$

which allows (11) to restore the correct marginal distribution $\mathcal{N}(\mathbf{X}_s; \mathbf{X}_0, \sigma^2_s)$. The subsequent noising process, which incorporates both deterministic and stochastic noise, ensures the transition to the correct noisy manifold [51].

The U-Net Jacobians provides deeper gradients, which can enhance reconstruction accuracy. It also provides a mechanism for imposing observation consistency in nonlinear inverse problems, where traditional gradient methods may not be feasible. Inspired by DPS [25], an augmentation of the solver with a gradient adjustment that maximizes the likelihood depending on time $p(\mathbf{Y}|\mathbf{X}_t)$ with respect to the observation is proposed. Specifically, the Jensen approximation is applied as

$$\begin{aligned} p\left(\mathbf{Y}|\mathbf{X}_t\right) &= \int p\left(\mathbf{Y}|\mathbf{X}_0\right) p\left(\mathbf{X}_0|\mathbf{X}_t\right)\, d\mathbf{X}_0 \\ &= \mathbb{E}_{p(\mathbf{X}_0|\mathbf{X}_t)}\left[p\left(\mathbf{Y}|\mathbf{X}_0\right)\right] \approx p\left(\mathbf{Y}|\mathbb{E}\left[\mathbf{X}_0|\mathbf{X}_t\right]\right) \\ &= p\left(\mathbf{Y}|\hat{\mathbf{X}}_{0|t}\right) \end{aligned} \tag{12}$$

where the final equality holds naturally due to the training objective in (6). Utilizing the approximation from (12), the correcting adjustment based on the Gaussian model becomes

$$\nabla_{\mathbf{X}_t} \log p\left(\mathbf{Y}|\mathbf{X}_t\right) \approx \nabla_{\mathbf{X}_t}\left\|\mathbf{Y} - \hat{\mathbf{X}}_{0|t} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\right\|_2^2. \tag{13}$$

Thus, the sampling process can be compactly expressed as

$$\begin{aligned} \mathbf{X}_s \leftarrow &\hat{\mathbf{X}}_{0|t} + \rho\nabla_{\mathbf{X}_t}\left\|\mathbf{Y} - \hat{\mathbf{X}}_{0|t} \times_1 \mathbf{P}_1 \times_2 \mathbf{P}_2\right\|_2^2 \\ &+ \alpha^2_{s|t}\left(\mathbf{X}_t - \hat{\mathbf{X}}_{0|t}\right) + \sigma_{s|t}\boldsymbol{\epsilon} \end{aligned} \tag{14}$$

where ρ is a step-size parameter for the gradient. The complete algorithm for DSB HSI super-resolution is summarized in Algorithm 1.

Unlike standard diffusion models that learn a generic mapping from unstructured noise to data, the DSB is designed for direct interpolation between two specific distributions. This makes the DSB a form of learned optimal transport, creating a direct and efficient pathway for reconstruction rather than an undirected generation from randomness. This targeted formulation confers critical advantages. Computationally, the bounded nature of the bridge—starting from the structured LR-HSI rather than pure noise—typically requires a much shorter sampling chain. This enhanced efficiency is vital for processing high-dimensional HSI cubes. More importantly, the DSB ensures superior spectral and spatial fidelity. The model's foundation in optimal transport theory constrains the solution space, promoting stable and deterministic outputs that reliably preserve the intricate spectral signatures essential for scientific analysis.

Furthermore, the DSB framework elegantly incorporates data consistency. During the reverse sampling process, intermediate clean image estimates can be explicitly constrained to match the observed HR-MSI data through gradient-based guidance. This fuses the powerful generative capability of diffusion models with the rigor of solving an ill-posed inverse problem, guaranteeing that the final HR reconstruction is both perceptually realistic and physically plausible. Consequently, the DSB emerges not just as a generative model but as a principled and robust solution for the exacting demands of HSI enhancement.

C. SDG DSB HSI Super-Resolution

Although the DSB can model the two distributions and introduce data consistency in the diffusion process, it failed to model the case of two inputs. In the HSI super-resolution, both LR-HSI and HR-MSI are observed and follow different degradation processes. Some methods take the additional input as conditions and connect it to the diffusion denoising network. However, the challenge remains in effectively integrating the two modalities (LR-HSI and HR-MSI) within the diffusion framework. Simply conditioning the diffusion denoising network on one of the inputs may not fully leverage the complementary information present in both datasets, which can be crucial for achieving high-quality super-resolution. Besides, the conditional diffusion model is not flexible, as it requires the input data pairs to be the same as the training data. However, the HSI and MSI pairs are very rare, and we cannot use the pretrained models that utilize a large amount of natural images. To solve these problems, we propose a SDG-DSB method for HSI super-resolution. Unlike previous methods that condition the diffusion process on one input (e.g., MSI or HSI), we introduce a novel approach that integrates both modalities directly into the diffusion process with posterior sampling. This allows for better utilization of the complementary information provided by the LR-HSI and HR-MSI, leading to higher quality super-resolution results. Besides, to better utilize the prior knowledge of natural images, we leverage the pretrained super-resolution Schrödinger Bridge. The flowchart of the proposed method is presented in Fig. 1. Next, we will describe the proposed SDG-DSB method in detail.

In the pretrained super-resolution Schrödinger Bridge, LR images and HR images are the two distributions that are learned. The pretrained model learns the denoising network with both the LR image and the HR image. In the generation process, the model can transform a LR image into a HR image. Ideally, the pretrained model can be used to enlarge the LR-HSI to the same spatial-resolution as HR-HSI since the pretrained model learns the spatial transform. However, in the pretrained model, the images are all natural RGB images, which have three channels. LR-HSI has hundreds of

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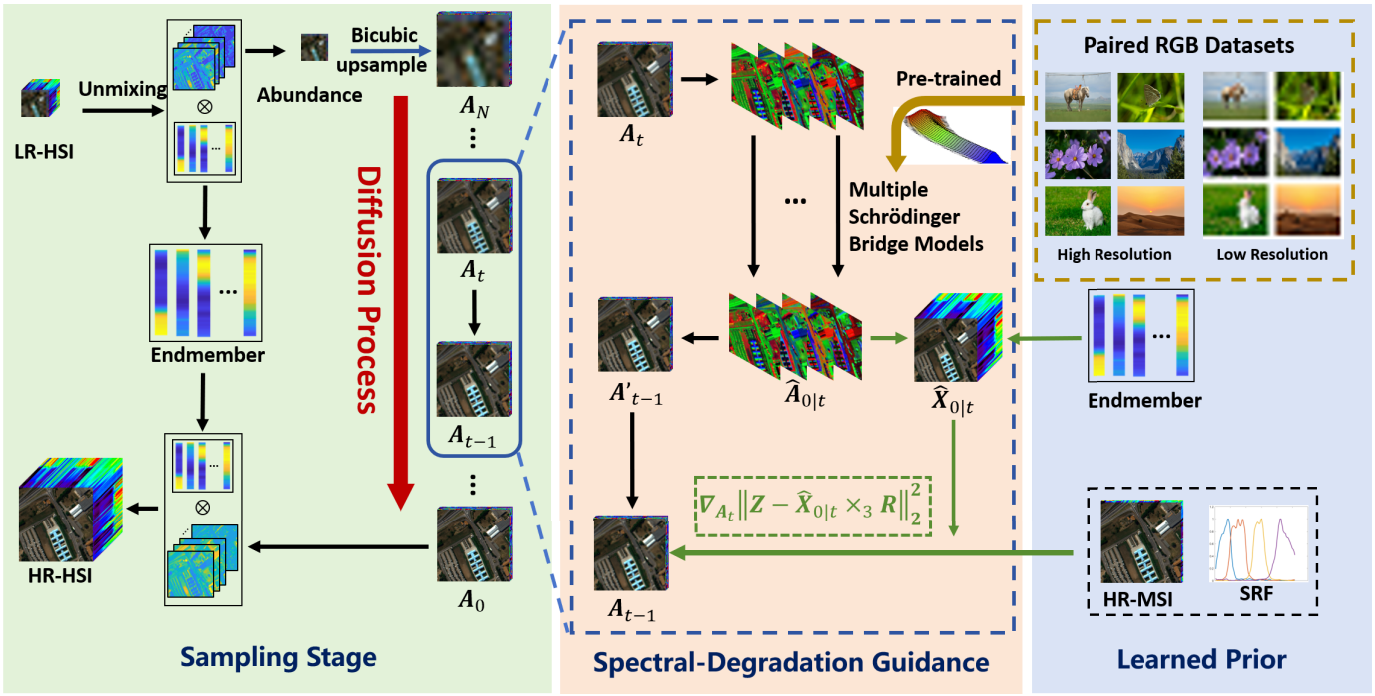


Fig. 1. Flowchart of the proposed method. The objective is to reconstruct a HR-HSI by fusing a LR-HSI with a HR-MSI image (e.g., RGB). The process begins by extracting a spectral prior from the LR-HSI and initializing a high-res estimate. This estimate is then iteratively refined through a generative diffusion SB model, learned from natural images. Crucially, at each step, the output is guided by a spectral-degradation guidance, ensuring the final result is both spatially enhanced and spectrally accurate.

channels and different sensors can generate different channel numbers. It is not possible to enlarge the LR-HSI with a pretrained model directly. In this article, we first use the linear unmixing model [52] that decomposes the LR-HSI into two parts: abundance maps and endmembers, as follows:

$$\mathbf{Y} = \mathbf{A} \times_3 \mathbf{E} + \mathbf{N} \tag{15}$$

where $\mathbf{A} \in \mathbb{R}^{h\times w\times r}$ is the abundance map, $\mathbf{E} \in \mathbb{R}^{L\times r}$ is the endmember matrix, and $\mathbf{N} \in \mathbb{R}^{h\times w\times L}$ is the noise matrix. In this article, we leverage Archetypal analysis (AA) [53], [54] as the core unmixing methodology to decompose the LR-HSI into endmembers and abundance maps. This approach was selected for its powerful geometric interpretation, which naturally identifies endmembers as convex combinations of spectral pixels located on the data simplex boundary, providing inherent robustness to spectral variability and eliminating the need for pure-pixel assumptions.

The number of endmembers r is typically much smaller than the number of spectral bands L, which indicates that the ground materials are limited and allows for dimensionality reduction and more efficient processing. LR-HSI contains high-resolution spectral information. Thus, we can reuse the endmember matrix $\mathbf{E}$ when reconstructing the HR-HSI. The abundance matrix $\mathbf{A}$ is spatially LR, which can be enlarged using the pretrained super-resolution DSB model. Although we have greatly reduced the number of the input's channels, the pretrained super-resolution DSB only accepts images that have three channels. Some methods [44] enforce the reduced channel number to be three, that is, $r = 3$. However, three channels are not always enough to fully explore the subspace and linear unmixing model. In this article, we propose using multiple generators to obtain multiple HR three-channel images. Then, we stack all the generated images along the channel dimension to get the final HR abundance tensor. The only constraint is that r should be a multiple of 3.

Mathematically, we first divide the abundance tensor into $\frac{r}{3}$ subtensors. We denote each subtensor as $\mathbf{A}^i \in \mathbb{R}^{h\times w\times 3}$, where $i = 1, 2, \ldots, r/3$. The abundance tensor $\mathbf{A}$ can be expressed as

$$\mathbf{A} = \text{stack}\left(\mathbf{A}^1, \mathbf{A}^2, \ldots, \mathbf{A}^{\frac{r}{3}}, \text{axis=3}\right). \tag{16}$$

Then, we feed the subtensors into $r/3$ pretrained super-resolution SB models.

According to [49], we first generate the estimated HR images

$$\hat{\mathbf{A}}^i_{0|t} \leftarrow G_{\theta^*}\left(\mathbf{A}^i_t\right) \tag{17}$$

where $G_{\theta^*}()$ is the pretrained neural network for natural image super-resolution.

Then, the HR abundance tensor can be obtained by

$$\hat{\mathbf{A}}_{0|t} = \text{stack}\left(\mathbf{A}^1_{0|t}, \mathbf{A}^2_{0|t}, \ldots, \mathbf{A}^{\frac{r}{3}}_{0|t}, \text{axis=3}\right) \in \mathbb{R}^{H\times W\times r}. \tag{18}$$

The initial estimated HR-HSI is obtained by

$$\hat{\mathbf{X}}_{0|t} \approx \hat{\mathbf{A}}_{0|t} \times_3 \mathbf{E}. \tag{19}$$

After obtaining the initial estimated HR-HSI, we can proceed to get final HR-HSI according to the remaining steps in Algorithm 1.

To this end, we only use the LR-HSI as input to generate the HR-HSI with the pretrained SB model. The HR-MSI is not fully explored and the spatial information is not injected into the fused HR-HSI. The CDDB-deep algorithm guarantees that

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Algorithm 2 SDG-DSB Algorithm

Require: G_{θ^*}, LR-HSI $\mathbf{Y}$, HR-MSI $\mathbf{Z}$, α_t, σ_t, $\alpha^2_{t-1|t}$, $\sigma^2_{t-1|t}$, ρ, r, β_1, β_2, $\mathbf{m}_t$, $\mathbf{v}_t$, ϵ

Decompose the LR-HSI $\mathbf{Y} \approx \mathbf{A} \times_3 \mathbf{E}$ using method [53]
Initialize $\mathbf{A}_N$ = upsample($\mathbf{A}$) using bicubic upsampling method, denote $\mathbf{A}_N$ = stack $\left(\mathbf{A}^1_N, \mathbf{A}^2_N, \ldots, \mathbf{A}^{\frac{r}{3}}_N, \text{axis=3}\right)$
Initialize $\mathbf{m}_N = \mathbf{v}_N = \mathbf{0}$
for $t = N$ **to** 1 **do**
  $\hat{\mathbf{A}}^i_{0|t} \leftarrow G_{\theta^*}(\mathbf{A}^i_t), i = 1, 2, \ldots, \frac{r}{3}$
  $\hat{\mathbf{A}}_{0|t} \leftarrow \text{stack}\left(\hat{\mathbf{A}}^1_{0|t}, \hat{\mathbf{A}}^2_{0|t}, \ldots, \hat{\mathbf{A}}^{\frac{r}{3}}_{0|t}, \text{axis=3}\right)$
  $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$
  $\mathbf{A}'_{t-1} \leftarrow (1 - \alpha^2_{t-1|t})\hat{\mathbf{A}}_{0|t} + \alpha^2_{t-1|t}\mathbf{A}_t + \sigma_{t-1|t}\boldsymbol{\epsilon}$
  $\hat{\mathbf{X}}_{0|t} \leftarrow \hat{\mathbf{A}}_{0|t} \times_3 \mathbf{E}$
  $\mathbf{g}_t \leftarrow \nabla_{\mathbf{A}_t} \|\mathbf{Z} - \hat{\mathbf{X}}_{0|t} \times_3 \mathbf{R}\|^2_2$
  $\mathbf{m}_{t-1}, \mathbf{v}_{t-1} \leftarrow \beta_1 \mathbf{m}_t + (1 - \beta_1)\mathbf{g}_t, \quad \beta_2 \mathbf{v}_t + (1 - \beta_2)\mathbf{g}^2_t$
  $\hat{\mathbf{m}}_{t-1}, \hat{\mathbf{v}}_{t-1} \leftarrow \frac{\mathbf{m}_{t-1}}{1-\beta_1^{N-t}}, \frac{\mathbf{v}_{t-1}}{1-\beta_2^{N-t}}$
  $\mathbf{A}_{t-1} \leftarrow \mathbf{A}'_{t-1} - \rho \frac{\hat{\mathbf{m}}_{t-1}}{\sqrt{\hat{\mathbf{v}}_{t-1}} + \epsilon}$
end for
return $\mathbf{X}_0 = \mathbf{A}_0 \times_3 \mathbf{E}$

the estimated clean image $\hat{\mathbf{X}}_{0|t}$ satisfies the observation consistency by introducing a gradient adjustment that minimizes the residual with respect to the observed LR-HSI. Since the LR-HSI is already leveraged as the input of the algorithm, reusing the LR-HSI to guide the sampling step cannot introduce other prior information. Thus, the improvement in generation is limited.

To address this problem, we propose a SDG sampling method that effectively utilizes both LR-HSI $\mathbf{Y}$ and HR-MSI $\mathbf{Z}$. Since both $\mathbf{Y}$ and $\mathbf{Z}$ are derived from the HR-HSI $\mathbf{X}$, we generate an estimated clean HR-HSI $\hat{\mathbf{X}}_{0|t}$ according to (8). Meanwhile, the generated clean HR-HSI should also meet the measurement condition of $\mathbf{Z} = \mathbf{X} \times_3 \mathbf{R}$ which can be achieved by taking the gradient adjustment minimizing the residual with respect to the observation $\mathbf{Z}$.

Thus, the variable that needs to be updated is the abundance tensor, and the sampling step can be represented as

$$\mathbf{A}_s \leftarrow \hat{\mathbf{A}}_{0|t} + \rho \nabla_{\mathbf{A}_t} \left\|\mathbf{Z} - \hat{\mathbf{X}}_{0|t} \times_3 \mathbf{R}\right\|^2_2 + \alpha^2_{s|t}\left(\mathbf{A}_t - \hat{\mathbf{A}}_{0|t}\right) + \sigma_{s|t}\boldsymbol{\epsilon}. \tag{20}$$

To accelerate convergence and reduce the number of sampling steps, we incorporate the Adam [55] optimization algorithm in updating $\mathbf{A}_t$. Adam adjusts learning rates using momentum terms derived from the first and second moments of the gradient, which smooth the update process and help adapt the step size. This approach reduces the runtime required for sampling while maintaining reconstruction accuracy. The detailed procedure can be found in Algorithm 2.

After obtaining the HR abundance tensor $\mathbf{A}_0$, we can reconstruct the HR-HSI by $\mathbf{X} = \mathbf{A}_0 \times_3 \mathbf{E}$. In this way, we can generate the HR-HSI from the LR-HSI and satisfy the spectral degradation process with respect to the HR-MSI. The proposed SDG sampling method fully exploits the LR-HSI, HR-MSI, and the prior probability distribution of natural images.

The complete algorithm for our SDG-DSB method is summarized in Algorithm 2.

IV. Experimental Analysis

In this section, we conduct comprehensive experiments on three remote sensing datasets to evaluate the effectiveness of the proposed method. We begin by briefly introducing the datasets and evaluation metrics used in the experiments. Following this, a detailed explanation of the experimental setup and procedures is provided. Next, we compare the performance of SDG-DSB against several state-of-the-art methods, presenting comparative data to emphasize the advantages of our approach. At last, we discuss the impact of key hyperparameters utilized in the experiments.

A. Datasets

In this article, we employed three widely used datasets for our experiments, namely.

1) Pavia University Dataset: The Pavia University dataset, acquired in 2003 using the ROSIS-03 airborne hyperspectral sensor, captures data over Pavia, Italy, across 115 spectral bands within the 0.43 to 0.86 μm wavelength range. After excluding 12 noisy bands, 103 spectral bands are typically utilized for analysis. From the original image, sized 610 × 340 × 103, a 256 × 256 × 103 patch was extracted to serve as the ground truth for processing and evaluation tasks. The LR-HSI is obtained through 4× downsampling using bicubic interpolation, while for the HR-MSI, we assume that the SRF of each multispectral band follows a Gaussian distribution. The HR-MSI is simulated by setting the mean of the Gaussian function to the specific central wavelength of each band.

2) Chikusei Dataset: The Chikusei dataset, captured using the Headwall Hyperspec-VNIR-C sensor, covers both agricultural and urban landscapes in Chikusei, Ibaraki, Japan. Spanning 128 spectral bands from 363 to 1018 nm, the dataset provides HR imagery with spatial dimensions of 2517 × 2335 pixels. A 256 × 256 pixel patch across all 128 spectral bands was selected from this imagery to serve as the ground truth for further analysis. The LR-HSI and HR-MSI are generated using the same method as for the Pavia dataset.

3) SanDiego Dataset: The SanDiego dataset, captured using the AVIRIS (airborne visible/infrared imaging spectrometer) sensor, provides detailed spectral information across 224 bands spanning wavelengths from 370 to 2510 nm. After excluding 22 noisy bands, a total of 202 effective bands were retained for analysis. The original image dimensions are 400 × 400 × 202. For experimental purposes, we extracted a 256 × 256 × 202 patch to serve as the ground truth for processing and evaluation tasks. The LR-HSI and HR-MSI are generated using the same method as for the Pavia dataset.

B. Evaluation Metrics

For the quantitative evaluation, we applied five well-known metrics: root-mean-squared error (RMSE), peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [56],

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TABLE I

COMPARISON RESULTS OF THE PROPOSED SDG-DSB AND SEVEN STATE-OF-THE-ART METHODS ON THE PAVIA DATASET. THE RUNTIME IS MEASURED IN SECONDS. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Method	Pavia Dataset					
	RMSE↓	PSNR↑	SAM↓	SSIM↑	UIQI↑	Time
CNMF	0.0128	38.76	3.40	0.973	0.994	6
HySure	0.0140	37.64	3.28	0.973	0.993	4
IRTenSR	0.0120	38.95	3.38	0.968	0.994	89
CuCaNet	0.0148	38.04	5.46	0.954	0.990	409
HyCoNet	0.0121	39.11	4.37	0.966	0.990	289
MIAE	0.0103	41.30	2.80	0.975	0.996	21
PLRDiff	0.0112	40.04	3.25	0.974	0.995	79
SDG-DSB	**0.0099**	**42.29**	**2.79**	**0.980**	**0.997**	25

TABLE II

COMPARISON RESULTS OF THE PROPOSED SDG-DSB AND SEVEN STATE-OF-THE-ART METHODS ON THE CHIKUSEI DATASET. THE RUNTIME IS MEASURED IN SECONDS. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Method	Chikusei Dataset					
	RMSE↓	PSNR↑	SAM↓	SSIM↑	UIQI↑	Time
CNMF	0.0287	38.91	5.12	0.966	0.988	6
HySure	0.0122	38.97	2.22	0.968	0.991	4
IRTenSR	0.0299	38.02	5.52	0.962	0.988	142
CuCaNet	0.0156	39.14	2.26	0.971	0.990	478
HyCoNet	0.0161	40.25	5.58	0.968	0.988	305
MIAE	0.0110	40.43	1.99	0.972	**0.993**	21
PLRDiff	0.0143	38.73	2.62	0.953	0.991	79
SDG-DSB	**0.0101**	**40.80**	**1.94**	**0.979**	**0.993**	25

TABLE III

COMPARISON RESULTS OF THE PROPOSED SDG-DSB AND SEVEN STATE-OF-THE-ART METHODS ON THE SANDIEGO DATASET. THE RUNTIME IS MEASURED IN SECONDS. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Method	SanDiego Dataset					
	RMSE↓	PSNR↑	SAM↓	SSIM↑	UIQI↑	Time
CNMF	0.0068	39.91	4.32	0.965	0.971	8
HySure	0.0071	39.50	4.68	0.963	0.983	5
IRTenSR	0.0070	40.17	4.70	0.949	0.984	193
CuCaNet	0.0068	40.47	4.46	0.969	0.984	599
HyCoNet	0.0061	41.98	4.25	0.964	0.978	386
MIAE	0.0060	41.33	3.99	0.966	0.985	22
PLRDiff	0.0066	40.66	4.94	0.971	0.978	79
SDG-DSB	**0.0057**	**42.24**	**3.86**	**0.975**	**0.986**	25

structural similarity index (SSIM) [57], and universal image quality index (UIQI) [58]. These metrics assess image quality from various perspectives, including noise levels, spectral accuracy, and structural integrity.

C. Settings

All DL experiments were conducted using PyTorch 2.1.0 on Python 3.10. The experiments were performed on a machine equipped with 64 GB of CPU memory and a GeForce GTX 4090 GPU. During the sampling process, we utilized an exponential noise schedule for the parameter β [59], with the decay rates for the first moment estimate β_1 and the second moment estimate β_2 in the Adam optimizer set to 0.9 and 0.999, respectively. The total number of sampling steps is set to 100, which will be discussed in Section IV-G2. To further enhance stability during the initial stages of sampling, a warm-up strategy was applied for the first few steps, facilitating faster convergence with the Adam optimizer.

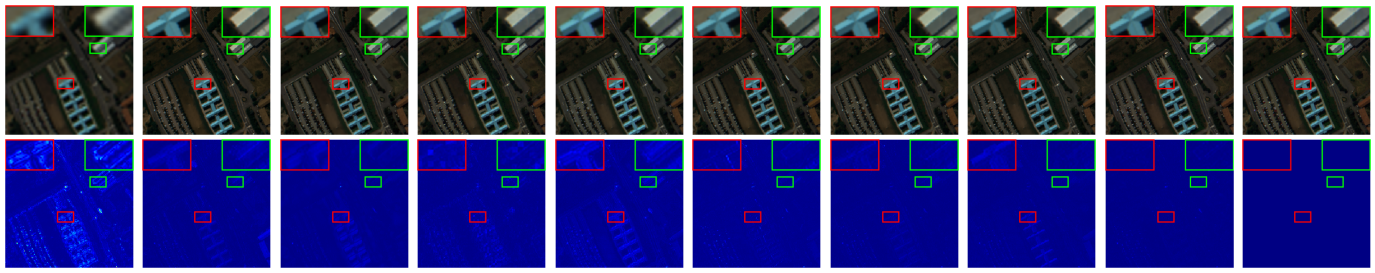
D. Experimental Results on Simulated Data

In this section, we conduct a thorough comparison of the proposed method against seven well-established approaches: CNMF [30], HySure [32], IRTenSR [60], CuCaNet [61], HyCoNet [62], MIAE [63], and PLRDiff [59]. Among these, CNMF, HySure, and IRTenSR represent traditional methods, while CuCaNet, HyCoNet, MIAE, and PLRDiff are unsupervised DL-based techniques. Notably, PLRDiff, similar to our proposed method, is grounded in diffusion model theory. To ensure a fair comparison in low-dimensional space, we maintained a consistent number of endmembers in methods involving subspace estimation across all experiments.

1) Quantitative Comparison: The experimental results for each method on the Pavia, Chikusei, and SanDiego datasets are summarized in Tables I–III, respectively. As depicted, the proposed method consistently surpasses all other approaches across five evaluation metrics, demonstrating superior super-resolution outcomes. Specifically, SDG-DSB achieves impressive PSNR values, which are significantly higher than those recorded by traditional methods like CNMF and HySure. These traditional methods, despite their relatively minimal runtime, yield super-resolution results that are approximately 2 dB lower than those of the proposed SDG-DSB method. In some instances, their performance is even less favorable, highlighting the limitations of these traditional approaches in delivering high-quality super-resolution results.

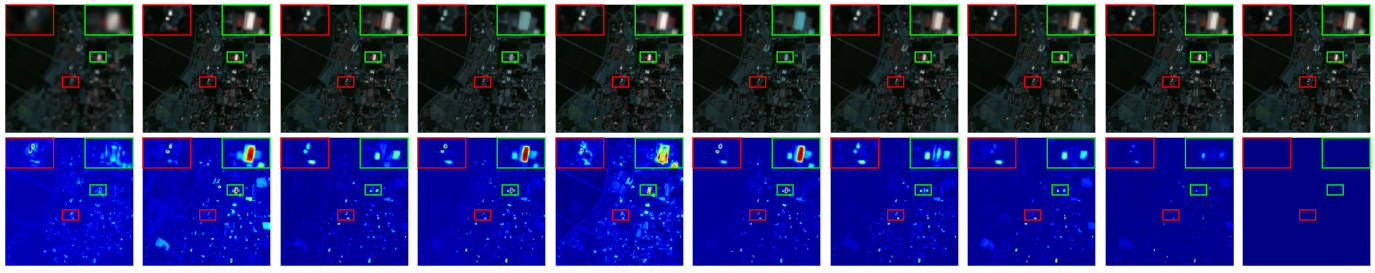
In the realm of DL methodologies, CuCaNet demonstrates occasional improvements over traditional methods; however, it is encumbered by excessively long runtimes, which limit its practical applicability. HyCoNet, while demonstrating enhanced super-resolution results, exhibits a runtime that is significantly longer, being at least ten times that of SDG-DSB. This extended processing time raises concerns regarding its efficiency in real-world scenarios where timely results are essential. MIAE, on the other hand, showcases commendable efficiency with a runtime of approximately 21 s. However, it still falls short in terms of accuracy across all datasets when compared to the proposed method, which demonstrates a more effective super-resolution capability. Furthermore, PLRDiff, another diffusion-based posterior sampling algorithm, produces acceptable super-resolution results, yet it lags behind SDG-DSB in both accuracy and runtime. The processing time for PLRDiff is approximately 54 s longer than that of the proposed method, reinforcing the advantage of our method not only in accuracy but also in efficiency.

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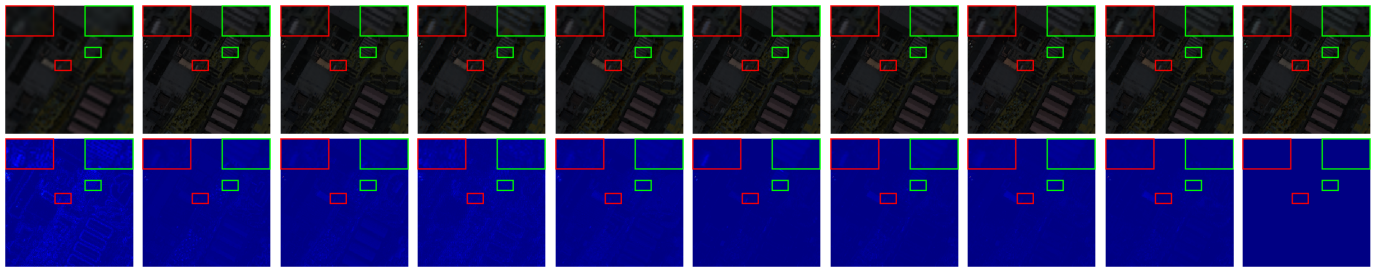
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 2. Visual results of different methods on Pavia dataset, where "GT" refers to the ground-truth image. The first row shows the RGB images (61–36–10 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference HR-HSIs. (a) LR-HSI ↑. (b) CNMF. (c) HySure. (d) IRTenSR. (e) CuCaNet. (f) HyCoNet. (g) MIAE. (h) PLRDiff. (i) SDG-DSB. (j) GT.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 3. Visual results of different methods on Chikusei dataset, where "GT" refers to the ground-truth image. The first row shows the RGB images (20–50–60 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference HR-HSIs. (a) LR-HSI ↑. (b) CNMF. (c) HySure. (d) IRTenSR. (e) CuCaNet. (f) HyCoNet. (g) MIAE. (h) PLRDiff. (i) SDG-DSB. (j) GT.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

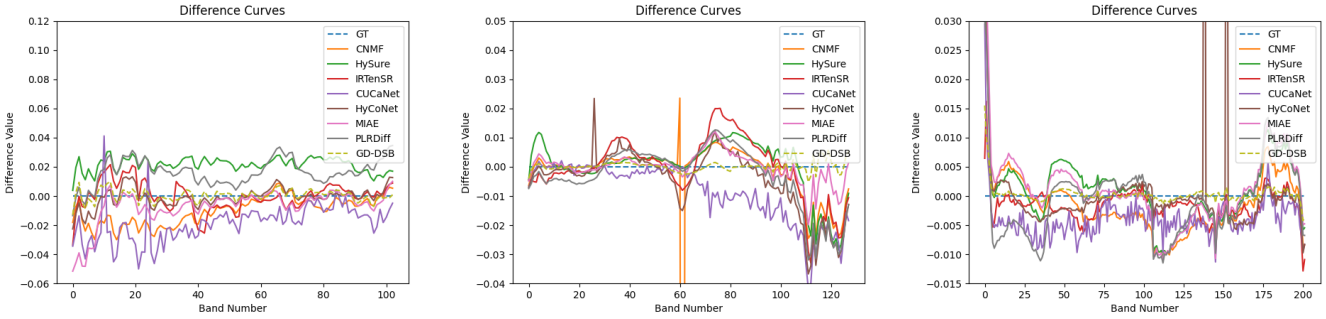
Fig. 4. Visual results of different methods on SanDiego dataset, where "GT" refers to the ground-truth image. The first row shows the RGB images (60–100–120 bands) of the estimated HR-HSIs, and the second row shows the difference images between the estimated and the reference HR-HSIs. (a) LR-HSI ↑. (b) CNMF. (c) HySure. (d) IRTenSR. (e) CuCaNet. (f) HyCoNet. (g) MIAE. (h) PLRDiff. (i) SDG-DSB. (j) GT.

Overall, the proposed method not only demonstrates a clear superiority in accuracy across the Pavia, Chikusei, and SanDiego datasets, but it also maintains a reasonably efficient runtime. This combination of high accuracy and acceptable processing times positions SDG-DSB as a leading solution in HSI super-resolution, highlighting its effectiveness and efficiency in comparison to both traditional and state-of-the-art DL methods. These findings underscore the potential of SDG-DSB for practical applications in HSI processing, where the demand for high-quality results is paramount.

2) Visual Comparison: To comprehensively evaluate the super-resolution results, we present a visual comparison between the proposed method and competing approaches. Figs. 2–4 showcase the RGB representations of the estimated HR-HSI alongside the corresponding difference images, which highlight the discrepancies between the estimated HR-HSI and the reference HR-HSI, across the Pavia, Chikusei, and SanDiego datasets. By examining the visual results, it becomes evident that traditional methods such as CNMF, HySure, and IRTenSR, while delivering reasonably good RGB reconstructions, exhibit significant reconstruction errors in the difference images. These errors indicate notable deviations from the ground truth, particularly in regions with fine spatial details. This is a clear limitation of these methods, which struggle to balance spatial and spectral fidelity.

Among the DL-based approaches, HyCoNet and MIAE demonstrate improved performance, achieving relatively smaller reconstruction errors when compared to traditional methods. These methods are able to better capture both spatial and spectral details, yielding more accurate super-resolution results. However, their difference images still reveal a greater degree of error than the proposed method, especially in

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Pavia (160, 120) Chikusei (60, 50) SanDiego (10, 100)

Fig. 5. Spectral difference between the ground truth and the HSIs reconstructed by all compared methods, where "GT" refers to the ground-truth image. Pixel positions for the Pavia, Chikusei, and SanDiego datasets are (160, 120), (60, 50), and (10, 100), respectively.

complex regions of the datasets. PLRDiff, another DL approach grounded in posterior sampling of diffusion models, produces visually appealing results but still falls short in terms of accuracy compared to SDG-DSB, with notable discrepancies in the difference images. The proposed method demonstrates a clear advantage in this visual analysis, producing the smallest reconstruction errors across all datasets. The difference images associated with SDG-DSB exhibit minimal deviation from the ground truth, reflecting its superior ability to reconstruct both the spatial structure and spectral characteristics of the HR-HSI. This highlights the method's robustness and accuracy in HSI super-resolution.

In addition to the visual comparison, Fig. 5 presents the spectral difference curves between the ground truth and the HSIs reconstructed by each method. These curves offer insight into the preservation of spectral information across the various approaches. The spectral difference curves for SDG-DSB consistently maintain values close to zero, indicating minimal spectral distortion. In contrast, other methods display significant deviations from the ground truth, particularly at certain wavelengths, implying higher levels of spectral distortion. This underscores the effectiveness of SDG-DSB in preserving spectral fidelity across different datasets, further affirming its superior performance in HSI super-resolution.

TABLE IV

AVERAGE QUANTITATIVE RESULTS ON DATASETS WITH THREE DIFFERENT BAND DISTRIBUTIONS (30 DB NOISE LEVEL). THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Dataset	Method	RMSE↓	PSNR↑	SAM↓	SSIM↑	UIQI↑
Pavia	CNMF	0.0131	38.12	3.46	0.967	0.994
	HySure	0.0147	37.26	3.33	0.968	0.992
	IRTenSR	0.0144	38.03	3.51	0.966	0.992
	CuCaNet	0.0153	37.82	5.27	0.951	0.986
	HyCoNet	0.0122	38.89	4.49	0.964	0.989
	MIAE	0.0112	40.41	3.25	0.970	**0.996**
	PLRDiff	0.0117	39.85	3.44	0.968	0.995
	SDG-DSB	**0.0107**	**41.23**	**3.08**	**0.974**	**0.996**
Chikusei	CNMF	0.0295	38.23	5.23	0.967	0.987
	HySure	0.0131	38.14	2.38	0.966	0.990
	IRTenSR	0.0315	37.72	5.29	0.962	0.987
	CuCaNet	0.0159	38.64	2.38	0.966	0.989
	HyCoNet	0.0168	39.36	5.19	0.968	0.987
	MIAE	0.0113	39.41	**2.00**	0.967	0.991
	PLRDiff	0.0130	37.36	2.47	0.958	0.990
	SDG-DSB	**0.0111**	**39.95**	2.05	**0.972**	**0.991**
SanDiego	CNMF	0.0070	39.15	4.52	0.963	0.970
	HySure	0.0076	38.94	4.78	0.963	0.981
	IRTenSR	0.0076	38.95	4.81	0.949	0.981
	CuCaNet	0.0071	39.68	4.61	0.964	0.982
	HyCoNet	0.0065	40.79	4.28	0.964	0.979
	MIAE	0.0064	40.40	4.17	0.966	0.984
	PLRDiff	0.0069	40.17	4.98	0.970	0.978
	SDG-DSB	**0.0060**	**41.52**	**4.08**	**0.972**	**0.985**

E. Experimental Results on Noisy and Blur-Degraded Simulated Data

To further verify the robustness of the proposed SDG-DSB model under more realistic sensor degradations, we conduct two additional experiments: 1) fusion under noisy conditions, and 2) fusion under Gaussian blur-based degradation. In the first setting, Gaussian noise with an SNR of 30 dB is added to the simulated LR-HSIs to emulate sensor noise such as shot and read noise. In the second setting, the LR-HSIs are generated by applying Gaussian point spread functions (PSFs) to the HR-HSIs before spatial decimation, simulating optical blur introduced by sensor modulation transfer functions (MTFs).

1) Results on Noisy Data: Table IV lists the quantitative results on three datasets under 30 dB noise. As expected, the presence of noise causes a slight decline in performance across all methods compared with the bicubic setting, indicating the increased difficulty of accurate spectral reconstruction. Nevertheless, SDG-DSB consistently achieves the best or second-best results on all metrics. For example, on the Pavia dataset, SDG-DSB attains an RMSE of 0.0107 and PSNR of 41.23 dB, surpassing the second-best method MIAE by 0.82 dB. On Chikusei, SDG-DSB obtains the lowest RMSE (0.0111) and the highest SSIM (0.972), demonstrating strong robustness to spectral noise. On SanDiego, SDG-DSB further achieves the best results on all five indicators, including a PSNR of 41.52 dB and SSIM of 0.972. These results suggest that the proposed diffusion-based model effectively suppresses noise propagation during sampling, while the adaptive residual guidance enhances denoising without compromising spectral fidelity.

2) Results on Gaussian-Blurred Data: Table V presents the results when the LR-HSIs are generated through Gaussian kernel degradation. Compared with the noisy case, Gaussian

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 6. Visual results of different methods on Houston dataset. (a) LR-HSI ↑. (b) HRMSI. (c) CNMF. (d) HySure. (e) IRTenSR. (f) CuCaNet. (g) HyCoNet. (h) MIAE. (i) PLRDiff. (j) SDG-DSB.

TABLE V

AVERAGE QUANTITATIVE RESULTS ON DATASETS WITH GAUSSIAN KERNEL DEGRADATION. THE BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Dataset	Method	RMSE↓	PSNR↑	SAM↓	SSIM↑	UIQI↑
Pavia	CNMF	0.0150	36.85	3.95	0.971	0.991
	HySure	0.0147	37.31	3.42	0.972	0.991
	IRTenSR	0.0123	38.69	3.52	0.968	0.993
	CuCaNet	0.0154	37.76	5.48	0.956	0.989
	HyCoNet	0.0122	39.07	4.44	0.967	0.990
	MIAE	0.0111	40.82	3.15	0.974	0.996
	PLRDiff	0.0114	40.19	3.25	0.975	0.995
	SDG-DSB	**0.0100**	**42.09**	**2.81**	**0.980**	**0.997**
Chikusei	CNMF	0.0265	36.81	4.52	0.965	0.988
	HySure	0.0134	38.02	2.61	0.964	0.990
	IRTenSR	0.0287	37.69	5.49	0.959	0.985
	CuCaNet	0.0162	38.03	2.68	0.963	0.989
	HyCoNet	0.0175	38.39	5.21	0.967	0.988
	MIAE	0.0136	38.55	2.42	0.967	0.991
	PLRDiff	0.0141	37.31	2.36	0.966	**0.992**
	SDG-DSB	**0.0117**	**38.98**	**2.17**	**0.974**	0.991
SanDiego	CNMF	0.0079	38.59	4.93	0.959	0.969
	HySure	0.0077	38.52	4.86	0.957	0.978
	IRTenSR	0.0079	39.16	4.93	0.944	0.977
	CuCaNet	0.0075	39.56	4.67	0.961	0.979
	HyCoNet	0.0071	39.93	4.63	0.963	0.972
	MIAE	0.0069	40.17	**4.57**	0.962	0.980
	PLRDiff	0.0119	36.99	5.33	0.954	0.968
	SDG-DSB	**0.0067**	**40.61**	4.60	**0.966**	**0.982**

blur introduces stronger spatial smoothing, making it harder to reconstruct sharp edges and detailed textures. Yet, SDG-DSB again achieves the best overall performance across datasets. Specifically, for Pavia, SDG-DSB yields an RMSE of 0.0100 and PSNR of 42.09 dB, clearly exceeding the runner-up MIAE (40.82 dB). For Chikusei, the proposed model reaches the lowest SAM (2.17), showing superior spectral preservation under blur. On SanDiego, it maintains the best PSNR (40.61 dB) and SSIM (0.966), confirming the model's ability to recover fine spatial–spectral details even under optical degradation.

Overall, the results across both degradation types verify that SDG-DSB maintains strong stability and generalization when facing noise or blur, outperforming existing fusion models under both spectral and spatial challenges.

F. Experimental Results on Real-Data

To further validate the effectiveness of the proposed method in real-world scenarios, we conducted experiments on the DFC2018 Houston dataset. This dataset, provided by Dr. Saurabh Prasad's laboratory at the University of Houston, consists of multisensor data, including HSIs, MSIs, and LiDAR data. The dataset contains 48 effective bands, covering the visible to near-infrared range. For the experiments, we extracted patches of sizes $64 \times 64 \times 48$ and $256 \times 256 \times 3$ to represent the HSI and MSI, respectively.

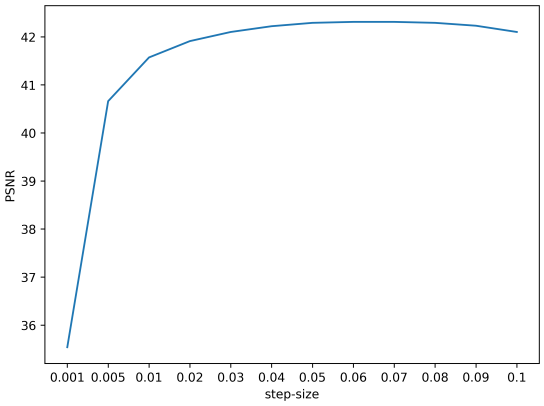


Fig. 7. Sensitivity analysis of the parameters ρ.

The reconstructed results of different methods are visualized as pseudo-color images in Fig. 6. As observed, the proposed method demonstrates superior performance in recovering fine details compared to the other methods. In particular, the HSIs generated by SDG-DSB exhibit reduced color distortion and better preservation of spatial and spectral consistency, making them more closely aligned with the real scene. These results highlight the proposed method's excellence in detail restoration and color balance, further confirming its superiority in real-world applications.

G. Parameter Analysis

1) Step-Size: In this section, we perform a sensitivity analysis on the step-size parameter ρ, as it plays a crucial role in the convergence and overall performance of the proposed method. As seen in (20), ρ controls the step-size for the gradient updates in the optimization process. Selecting an inappropriate value for ρ can significantly impact reconstruction accuracy. If ρ is set too low, the method may converge slowly or fail to reach the optimal solution, as the updates will be too small to make meaningful progress. Conversely, if ρ is set too high, the method risks overfitting, where it becomes overly sensitive to the target, leading to instability in the reconstruction results.

Fig. 7 illustrates the impact of varying ρ on the PSNR of the reconstructed HR-HSI using the Pavia dataset. From the figure, it is evident that the PSNR values increase as ρ approaches 0.05, indicating that this is the optimal step-size for the best reconstruction performance. Thus, $\rho = 0.05$ strikes the right balance between convergence and avoiding overfitting, allowing the method to achieve superior results in HSI super-resolution. This optimal setting not only improves the

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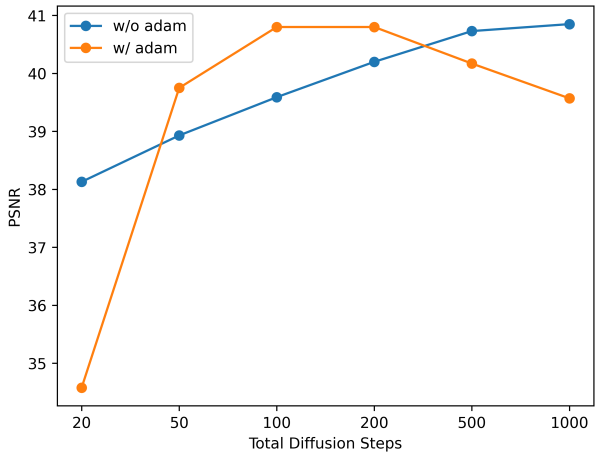
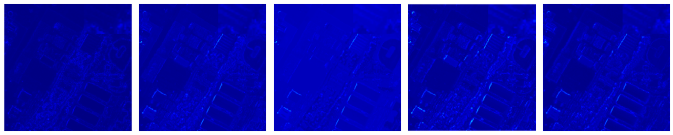


Fig. 8. Performance of SDG-DSB under different total numbers of diffusion steps.



(a) (b) (c) (d) (e)

Fig. 9. Heat maps of SAM on SanDiego dataset. (a) LR-HSI ↑. (b) Rank = 3. (c) Rank = 6. (d) Rank = 9. (e) Rank = 12.

super-resolution accuracy but also contributes to the stability and efficiency of the proposed method.

2) Total Diffusion Steps: The total number of diffusion steps is a key factor influencing both the performance and runtime of the proposed SDG-DSB method, especially when leveraging large pretrained model weights. A balance must be struck between the number of diffusion steps and the resulting super-resolution accuracy. Fig. 8 illustrates the performance trends under varying diffusion steps for both "w/o Adam" and "w/ Adam" cases.

In the "w/o Adam" scenario, increasing the number of diffusion steps leads to improved super-resolution accuracy, but the marginal performance gains diminish as the number of steps increases. Additionally, the longer runtime associated with more diffusion steps becomes a limiting factor, especially when the improvement in performance plateaus. In contrast, when "w/ Adam" is incorporated, the method exhibits rapid convergence, achieving peak performance at around 100 diffusion steps. After 200 steps, however, performance begins to decline, likely due to overfitting caused by excessive sampling. This indicates that, while Adam accelerates convergence and reduces the need for a large number of diffusion steps, exceeding the optimal range can negatively impact the reconstruction quality.

To strike a balance between performance and runtime efficiency, we adopt the "w/ Adam" approach and set the total number of diffusion steps to 100. This setting ensures both high super-resolution accuracy and a reasonable computational cost, making the method practical for real-world applications.

3) Rank Setting for the Subspace: In (15) and (16), the LR-HSI is decomposed into the product of abundance maps $\mathbf{A}$ and endmembers $\mathbf{E}$, where $\mathbf{A}$ is formed by stacking several three-channel tensors. Therefore, the parameter "r," representing the rank of the $\mathbf{A}$ and $\mathbf{E}$, must be a multiple of 3. In Table VI, we present the quantitative results on the SanDiego dataset with different rank settings. The performance shows a clear upward trend as the rank increases from 3 to 6, where both PSNR and SSIM improve significantly, indicating enhanced spatial detail recovery and spectral reconstruction quality. However, when the rank further increases to 9 and 12, the PSNR slightly decreases and the SAM increases, suggesting that over-parameterization makes the optimization of abundance and endmember tensors more unstable. This observation implies that an excessively large rank may lead to redundant spectral bases and local inconsistency among spectral groups. To visually confirm this effect, the per-pixel SAM heat maps in Fig. 9 illustrate the spatial distribution of spectral errors. The heat maps reveal that rank = 6 yields the most uniform and lowest SAM values across the scene, while higher ranks (e.g., 9 and 12) introduce mild spectral variations around object boundaries. These results jointly demonstrate that a moderate rank not only achieves superior quantitative performance but also maintains inter-group spectral consistency, avoiding spectral seams and ensuring stable reconstruction.

TABLE VI
RESULTS UNDER DIFFERENT RANK SETTING. THE RUNTIME IS MEASURED IN SECONDS

Rank	3	6	9	12
PSNR	34.53	42.24	41.78	41.29
SAM	9.54	3.86	4.01	4.11
Time	13	25	38	49

V. CONCLUSION

In conclusion, this article presents an innovative unsupervised framework for HSI super-resolution using the SDG DSB. By addressing the limitations of conventional methods, which seldom leverage prior knowledge from other images, this approach enhances both spatial and spectral resolution in fused images. The SDG-DSB effectively transforms LR-HSI into HR-HSI while maintaining data consistency with HR-MSI. The model avoids the dependence on arbitrary weighting parameters, improving the super-resolution quality, and leverages pretrained models to overcome challenges associated with limited training data. Comprehensive experiments on multiple datasets demonstrate the superiority of this method over existing super-resolution techniques, confirming its potential for advancing remote sensing applications.

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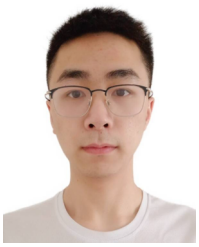
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